

ENGINEERING RELEASE GUIDE

# Anticipy Hardware Handoff

## Custom PCB EVT R0B

*One controlled package for electrical, mechanical, firmware, manufacturing and final assembly*

**Release ANT-PROD-R0B-EVT-HANDOFF-2026-08-29**

Prepared August 29, 2026

### **FABRICATION BLOCKED**

The source is editable and useful, but 176 PCB connections remain unrouted. The battery, final enclosure and production firmware are also not released.

Release: ANT-PROD-ROB-EVT-HANDOFF-2026-08-29

## 1. What this package is

This package gives a senior electrical engineer, mechanical designer, firmware engineer, PCBA factory, and final assembler one controlled source of truth for turning the Anticipy custom-PCB concept into working EVT units.

It includes the editable KiCad project, custom footprints, draft BOM, pin map, verified board outline, provisional board-only STEP/DXF, engineering scope, manufacturing RFQs, firmware requirements, assembly plan, QA gates, and ready-to-send messages.

It does not pretend unfinished items are finished. The board is not fabrication-ready, the battery is not selected, and the final enclosure CAD and firmware binary do not exist yet.

## 2. Plain-English definitions

- **PCB:** the bare green circuit board.
- **BOM:** bill of materials, meaning the shopping list of electronic parts.
- **PCBA:** the PCB after the electronic parts have been soldered onto it.
- **Schematic:** the electrical map showing what connects to what.
- **PCB layout:** the physical placement and copper traces on the board.
- **Gerbers:** factory images of every PCB layer.
- **Drill files:** instructions for every plated hole, slot, and non-plated cut.
- **CPL or pick-and-place file:** the X/Y position, rotation, and side for every machine-placed part.
- **STEP:** a 3D CAD model used for collision and enclosure design.
- **DXF:** a 2D cutting profile, often used for sheet metal or outline transfer.
- **EVT:** engineering validation test, the first hardware revision used to prove the design works.

The PCB is not the BOM. The BOM is not the enclosure. The PCBA factory does not automatically create a finished pendant.

## 3. Frozen product intent

Anticipy is a camera-free wearable Bluetooth audio pendant. The device is intended to record ambient audio, store audio when offline, transfer live and stored data to an iPhone, indicate state using a tiny red/blue light, provide haptic feedback, detect motion/wear state, and support secure reset and factory programming.

| Requirement       | Frozen target                                                                      |
|-------------------|------------------------------------------------------------------------------------|
| Finished exterior | 51.0 x 21.0 x 11.0 mm maximum                                                      |
| Shape             | Rounded capsule with proportions similar to PLAUD NotePin S; no copied private CAD |
| Mass              | Target under 20 g                                                                  |

| Requirement          | Frozen target                                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------|
| Centre structure     | Polycarbonate                                                                                            |
| Front and rear faces | 0.60 mm brushed 5052-H32 aluminum                                                                        |
| Radio region         | Polycarbonate only over antenna; no metal, battery, motor, magnet, chain hardware, or conductive coating |
| Camera               | None                                                                                                     |
| PCB                  | 45.8 x 18.0 mm current outline; 4-layer; 0.80 mm production target                                       |

## 4. Current electrical architecture

The ROB design uses:

- Raytac MDBT50Q-1MV2 nRF52840 radio/processor module.
- Nordic nPM1300 charger and power-management IC.
- Two Same Sky PDM microphones.
- Macronix 4-Gbit SLC serial NAND.
- ST LIS2DW12 accelerometer.
- USB-C with data, CC, and VBUS protection.
- Red/blue status LED, user button, haptic driver, factory pogo pads, battery lands, and motor lands.

The battery is a placeholder only. The required pack must be protected, three-wire, include a 10 kohm NTC, and fit within a maximum installed envelope of 26.0 x 15.5 x 4.8 mm including protection, insulation, folded tabs, tolerance, and swelling allowance.

## 5. Honest engineering status

The automated ROB verifier reports:

- 4 copper layers.
- 45.8 x 18.0 mm geometric outline.
- 70 electrical references.
- 223 source-of-truth pins checked.
- 0 geometric DRC violations.
- 176 unrouted PCB connections.**
- 0.50 mm provisional battery-to-antenna gap.
- 2.00 mm conservative motor-to-antenna gap.

Therefore, this is a strong starting package, not a factory release.

## 6. Who does what

### Senior PCB engineer

The senior PCB engineer owns the electrical design until a release package exists. They must review every symbol and footprint, reconcile the 0.80/0.82 mm thickness issue, select or approve the exact battery, manually complete routing, review the nPM1300 loops against Nordic's reference, obtain the fabricator's impedance stack-up, run final ERC/DRC/net-equivalence checks, create a PCBA STEP, generate manufacturing outputs, and support first-board bring-up.

### PCBA factory

The factory supplies or fabricates the bare PCB, sources approved SMT parts, prints solder paste, places components, reflows, inspects, programs/tests if quoted, and returns assembled PCBAs. The factory should build exactly two first articles and hold the other ten until written release.

### Mechanical CAD engineer

The mechanical engineer designs the polycarbonate centre, aluminum faces, button actuator, light pipe, microphone acoustic wells, USB opening, battery retention, motor retention, chain/bail interface, sealing, and assembly strategy around the released PCBA STEP and exact battery STEP.

### Final integration technician

The final technician receives tested PCBAs, batteries, motors, enclosure parts, firmware, assembly drawings, adhesives, insulation, acoustic mesh, and QA forms. They install the rear hand-solder parts, insulate and strain-relieve them, close the enclosure, cure adhesives, and run sealed-unit tests.

## 7. What each recipient receives

### PCB engineer package

- Native KiCad `.kicad_pro`, `.kicad_sch`, `.kicad_pcb`, and `.kicad_dru`.
- Custom `.pretty` footprint library and `fp-lib-table`.
- Draft BOM, pin map, pad/net report, verifier, design generator, requirements, placement plan, stack-up request, and current DRC report.
- Nordic nPM1300 reference-layout archive.
- Exact scope of work and release checklist.

### Mechanical CAD package

- Product envelope and material strategy.
- Provisional board-only STEP and board-outline DXF.
- PCBA placement image and interface requirements.
- Required CAD deliverable list.

The provisional files are for early packaging studies only. The final enclosure must use the engineer-released PCBA STEP and exact battery STEP.

### PCBA factory package

Before release, the factory may receive only the RFQ and stack-up request. After release, it receives one locked ZIP containing Gerbers, drills, BOM, CPL, assembly drawings, fab drawing, stack-up, IPC-356, schematic PDF, PCBA STEP, programming instructions, firmware hash, and first-article test plan.

### Final assembly package

- Released PCBA and battery STEP files.
- Complete enclosure STEP and exploded STEP.
- Polycarbonate production STEP/3MF.
- Aluminum front/rear/markings DXFs.
- Mechanical BOM.
- Adhesive and cure specification.
- Assembly traveler.
- Firmware and test instructions.
- Per-unit serial and QA record.

## 8. Materials and who supplies them

The pendant is a sandwich:

1. A 0.60 mm brushed 5052-H32 aluminum front face.
2. A polycarbonate centre frame that holds and electrically isolates the hardware.
3. The tested PCBA, protected battery, motor, insulation, foam, light pipe, and acoustic mesh.
4. A 0.60 mm brushed 5052-H32 aluminum rear face.
5. A polymer-only radio zone over the antenna.

You do not normally ship raw aluminum to the PCB engineer. The engineer specifies interfaces. The metal shop normally supplies certified aluminum sheet and cuts it from released DXFs. The PCBA factory supplies PCB laminate, copper, solder mask, finish, stencil, paste, and approved SMT parts unless the purchase order says Anticipy will consign a part. Anticipy should consign the exact approved battery and haptic motor for the first batch.

## 9. Seven-day rush attempt

The seven-day clock is a target, not a promise.

| Day   | Required outcome                                                                   |
|-------|------------------------------------------------------------------------------------|
| Day 0 | Hire PCB lead; send RFQs; freeze decision owner; select battery candidate; reserve |

| Day   | Required outcome                                                                                                                          |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------|
|       | factory capacity                                                                                                                          |
| Day 1 | Complete schematic/footprint review, routing, stack-up negotiation, mechanical interface review, and release-candidate checks             |
| Day 2 | Release manufacturing package only if every gate passes; factory starts two first articles; mechanical CAD continues around released PCBA |
| Day 3 | Fabrication/assembly; firmware bench image and pogo fixture ready; enclosure prototypes printed                                           |
| Day 4 | Bring up FA-001 and FA-002; measure rails; program; test USB, BLE, microphones, storage, motion, LED, and haptic                          |
| Day 5 | If both pass, release remaining ten PCBAs; freeze enclosure and cut/print parts                                                           |
| Day 6 | Integrate and seal first two complete pendants; run acoustic/RF/charge/runtime tests                                                      |
| Day 7 | Finish remaining units only if the first two sealed units pass; serialize, clean, and package                                             |

One to five working EVT units in a week may be possible with immediate engineering and a first-spin success. Ten polished sealed units in one week cannot be guaranteed.

## 10. Release sequence

The only safe sequence is:

1. Engineering release candidate.
2. Zero unconnected items and zero unresolved DRC errors.
3. Fabricator-reviewed 0.80 mm stack-up and 90-ohm USB geometry.
4. Exact battery approved.
5. Manufacturing package checksummed and signed.
6. Two first articles manufactured.
7. Both pass Gate A bench testing.
8. Written **RELEASE REMAINING TEN** approval.
9. Two sealed first articles pass Gate B.
10. Customer release.

## 11. Immediate owner actions

- Send the complete ZIP to one accountable senior PCB engineer.

- Send the factory RFQ immediately, but do not issue fabrication files or a purchase order.
- Make battery selection the first engineering decision.
- Assign one mechanical CAD owner and one firmware/bring-up owner.
- Reserve access to a current-limited bench supply, oscilloscope, DMM, SWD programmer, USB test host, BLE phone, audio stimulus, and pogo fixture.
- Use the project-control workbook to record one owner and deadline for every open gate.

## 12. What is deliberately missing

The following files cannot honestly be supplied yet because the design has not reached that stage:

- Released Gerbers and drills.
- Released BOM and CPL.
- Final PCBA STEP with all components.
- Exact battery STEP and approved battery MPN.
- Final enclosure STEP, production 3MF, and aluminum DXFs.
- Production firmware binary and SHA-256.
- Final programming fixture drawing and test software.

Their required filenames and acceptance rules are included so the responsible engineer knows exactly what must be produced.

APPENDIX A

# Open items and decisions



## P0 - blocks fabrication

| Decision/item         | Required evidence                                                                                             | Owner                 | Status               |
|-----------------------|---------------------------------------------------------------------------------------------------------------|-----------------------|----------------------|
| Exact battery         | MPN, max drawing/STEP, protection, 10k NTC, polarity, limits, UN38.3, IEC 62133 evidence, MSDS, sample, stock | PCB lead + Anticipy   | Open                 |
| Complete PCB routing  | Zero unconnected items and reviewed routes                                                                    | PCB lead              | 176 open connections |
| PMIC review           | Schematic and layout signed against Nordic reference                                                          | PCB lead              | Open                 |
| 0.80 mm stack-up      | Fabricator drawing, Dk, copper, dielectric, USB geometry                                                      | PCB lead + factory    | Open                 |
| Thickness discrepancy | Explain/resolve 0.80 mm target versus 0.82 mm board-only STEP                                                 | PCB lead + mechanical | Open                 |
| Final PCBA STEP       | All maximum-height components and interface bodies                                                            | PCB lead              | Open                 |
| Factory package       | Gerbers, drills, IPC-356, BOM, CPL, drawings, checksums                                                       | PCB lead              | Open                 |

## P0 - blocks first-article bring-up

| Decision/item               | Required evidence                                             | Owner          | Status |
|-----------------------------|---------------------------------------------------------------|----------------|--------|
| Production/factory firmware | Reproducible source, binary, hash, flash/verify commands      | Firmware lead  | Open   |
| NAND plan                   | ECC, bad blocks, filesystem, recovery and secure erase proven | Firmware lead  | Open   |
| Pogo fixture                | Drawing, pin map, programmer and test connections             | PCB + firmware | Open   |
| Numeric test limits         | Rail, ripple, current, audio, RF, thermal and runtime limits  | PCB + firmware | Open   |

## P1 - blocks sealed units

| Decision/item           | Required evidence                                    | Owner                  | Status |
|-------------------------|------------------------------------------------------|------------------------|--------|
| Final enclosure         | STEP, 3MF, DXFs, drawings, mechanical BOM            | Mechanical lead        | Open   |
| Acoustic system         | Mesh, port geometry, adhesive keepout, test limits   | Mechanical + audio     | Open   |
| Antenna system          | Polymer-only region and sealed-unit RF acceptance    | PCB + mechanical       | Open   |
| Adhesive/seal process   | Exact product, pattern, cure, rework and inspection  | Mechanical + assembler | Open   |
| Final integration owner | Named local technician, equipment, schedule and cost | Anticipy               | Open   |

## Decision log template

For every choice, record:

- Decision ID and date/time.
- Exact part/file/revision.
- Options considered.
- Evidence and test result.
- Decision and owner.
- Downstream files that must be updated.
- Whether a new revision/checksum is required.

APPENDIX B

# Seven-day rush plan

## Goal

Attempt to produce 12 custom-PCB EVT assemblies, with two first articles built and tested before the remaining ten. The practical success target is one to five working units in seven days. Ten sealed and polished units remain schedule-risky because revision-one PCB, battery, firmware, and enclosure work are still open.

## Staffing

- One accountable senior PCB engineer.
- One firmware/bring-up engineer.
- One mechanical CAD engineer.
- One PCBA factory project owner.
- One final integration technician; two stations preferred after first-article release.
- One Anticipy decision owner who can approve or stop work immediately.

## Schedule

| Day | PCB/electrical                                                       | Mechanical                                               | Firmware/test                                             | Manufacturing/integration                                   |
|-----|----------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------|
| 0   | Engineer accepts scope; review blockers                              | Start packaging study                                    | Freeze bring-up plan                                      | Send RFQs; reserve capacity                                 |
| 1   | Battery decision; footprint review; complete routing; DRC/net checks | Develop centre/face concepts around provisional geometry | Create minimal bench image, self-tests, SWD plan          | Factory returns stack-up, geometry, stock and schedule      |
| 2   | Release only if all Gate 0 checks pass                               | Update from released PCBA STEP and battery               | Release hashed FA firmware/test instructions              | Start two first articles; print enclosure prototypes        |
| 3   | Support DFM/CAM and answer exceptions                                | Fit/tolerance review                                     | Prepare fixture, scripts, iPhone/BLE tools                | Fabricate/assemble FA-001/FA-002                            |
| 4   | Lead current-limited bring-up                                        | Correct mechanical issues found                          | Test rails, USB, BLE, audio, storage, motion, LED, haptic | Factory supplies inspection evidence                        |
| 5   | Close faults or issue remaining-ten release                          | Freeze enclosure after FA evidence                       | Freeze passed firmware hash                               | Assemble remaining ten only if both FAs pass                |
| 6   | Support integration faults                                           | Produce/finish parts                                     | Sealed-unit regression                                    | Integrate two sealed first articles                         |
| 7   | Final disposition                                                    | Final mechanical corrections                             | Overnight/runtime/backfill review                         | Integrate remaining passed units, serialize, clean, package |

## Daily decision gates

- End of Day 1: Can the board be released without waivers?
- End of Day 2: Did the factory accept the exact technology and files?
- End of Day 4: Did both open-board first articles pass?
- End of Day 6: Did both sealed first articles pass?

If any answer is no, preserve the hardware and continue the EVT correction loop. Do not convert an engineering failure into a pass to protect the calendar.

APPENDIX C

# File dictionary

## Files present now

| File/type                            | Meaning                                     | Primary user             | Status                      |
|--------------------------------------|---------------------------------------------|--------------------------|-----------------------------|
| .kicad_pro                           | KiCad project settings                      | PCB engineer             | Editable WIP                |
| .kicad_sch                           | Electrical schematic                        | PCB engineer/firmware    | Editable WIP                |
| .kicad_pcb                           | Board outline, placement and future routing | PCB engineer             | Editable WIP, unrouted      |
| .kicad_dru                           | PCB design rules                            | PCB engineer             | WIP                         |
| .pretty/<br>*.kicad_mod              | Custom footprints                           | PCB engineer/assembler   | Must be verified            |
| BOM_DRAFT_DO<br>_NOT_PRODUCE.<br>csv | Draft part list                             | PCB engineer/procurement | Not procurement-released    |
| pad_net_report.csv                   | Pin/net cross-check data                    | PCB engineer             | Engineering reference       |
| verify_project_r0b.py                | Structural/release verifier                 | PCB engineer             | Reports 176 unrouted items  |
| placement.png/.svg                   | Visual component placement                  | All engineering roles    | Provisional                 |
| PROVISIONAL_BOARD_ONLY.step          | 3D bare-board solid only                    | Mechanical engineer      | Provisional, 0.82 mm export |
| PROVISIONAL_BOARD_OUTLINE.dxf        | 2D board outline                            | Mechanical engineer      | Provisional                 |

## Files that must be created before fabrication

| Required file                 | What it does                          | Creator      |
|-------------------------------|---------------------------------------|--------------|
| Release README with checksums | Locks revision and file set           | PCB engineer |
| Gerber ZIP                    | Defines every manufactured PCB layer  | PCB engineer |
| PTH/NPTH drill ZIP            | Defines holes and slots               | PCB engineer |
| Fabrication drawing           | Defines material, finish, tolerances, | PCB engineer |

| Required file            | What it does                                          | Creator                  |
|--------------------------|-------------------------------------------------------|--------------------------|
|                          | impedance and notes                                   |                          |
| Approved stack-up        | Defines real four-layer construction and USB geometry | Factory + PCB engineer   |
| Released BOM             | Controls procurement and substitutions                | PCB engineer/procurement |
| CPL CSV                  | Controls placement X/Y/rotation/side                  | PCB engineer             |
| Assembly drawings        | Controls orientation and special process              | PCB engineer             |
| IPC-356                  | Electrical netlist for CAM/test                       | PCB engineer             |
| PCBA STEP                | Enables component collision review                    | PCB engineer             |
| Programming/test package | Controls firmware and factory test                    | Firmware + PCB engineer  |

### Files that must be created before enclosure/tooling release

| Required file                    | What it does                                     | Creator                      |
|----------------------------------|--------------------------------------------------|------------------------------|
| Exact battery STEP/drawing       | Controls battery fit and safety                  | Battery supplier + engineer  |
| Enclosure assembly/exploded STEP | Controls complete mechanical fit                 | Mechanical engineer          |
| Polycarbonate centre STEP/3MF    | Controls structural centre                       | Mechanical engineer          |
| Aluminum front/rear DXF          | Controls sheet cutting                           | Mechanical engineer          |
| Marking DXF                      | Controls ANTICIPY mark                           | Mechanical engineer          |
| Mechanical drawing/BOM           | Controls materials, tolerances and parts         | Mechanical engineer          |
| Assembly traveler                | Controls sequence, adhesive, cure and inspection | Mechanical + final assembler |

### What to send when

- Send the entire ZIP now to the accountable PCB engineer.
- Send the master guide, mechanical folder and placement files now to the mechanical engineer.
- Send RFQs now to factories for capacity review only.



- Send final manufacturing outputs only after Gate 0 is signed.
- Send final assembly outputs only after both open-board first articles pass.

APPENDIX D

# PCB engineer scope

## Assignment

Take **ANT-PROD-ROB** from engineering work-in-progress to an EVT fabrication release for 12 PCBAs, with a mandatory hold after two first articles.

## Starting point

- Editable KiCad schematic and PCB.
- Custom footprint library.
- 45.8 x 18.0 mm outline.
- Four-layer 0.80 mm target stack.
- 70 references and 223 checked design pins.
- 176 unrouted PCB connections.
- Draft BOM, pin map, requirements, placement plan, and Nordic reference archive.

## Required work

### Electrical review

- Review every schematic symbol, pin number, power pin, no-connect, polarity, and net label against the current manufacturer datasheet.
- Verify every footprint land pattern, exposed pad, paste aperture, maximum height, microphone port, USB shell slot, LED orientation, and module antenna keepout.
- Confirm the nPM1300 configuration, NTC circuit, buck startup values, charge limits, high-current grounds, and net ties against Nordic's reference.
- Confirm USB-C CC behavior, ESD topology, VBUSOUT connection, 27-ohm series resistors, and D+/D- polarity.
- Confirm the raw NAND choice is acceptable for the firmware schedule. If not, propose a lower-risk soldered storage part with a documented size, power, and firmware tradeoff.

### Mechanical and sourcing decisions

- Select or approve the exact protected three-wire battery before release.
- Obtain its maximum-tolerance STEP/drawing, protection details, NTC curve, polarity, voltage/current/temperature limits, UN38.3, IEC 62133 evidence, MSDS, lot traceability, and stock confirmation.
- Reconcile the KiCad 0.80 mm target with the board-only STEP export that measures 0.82 mm.
- Review the battery, motor, USB, button, LED, microphone, pogo, and antenna interfaces with the mechanical engineer.

### Layout completion

- Preserve an uninterrupted L2 ground reference.
- Keep all copper, components, and vias out of the Raytac all-layer antenna keepout.

- Route and review every connection manually; do not accept an autorouter result without net-by-net review.
- Keep nPM1300 switch loops and bypass paths compact and faithful to Nordic guidance.
- Route USB D+/D- as a controlled 90-ohm differential pair using the fabricator-approved stack-up and geometry.
- Review return paths, stitching, current density, via sizes, thermal reliefs, exposed pads, acoustic clearance, ESD placement, and test access.
- End with zero unconnected items and zero unresolved DRC errors.

## Release outputs

Produce:

1. Complete editable KiCad project.
2. Final released schematic and PCB.
3. Footprint and symbol verification record.
4. ERC, DRC, net-equivalence, and unconnected-item reports.
5. Gerber X2/RS-274X ZIP.
6. Separate PTH and NPTH Excellon drill ZIP plus drill map.
7. IPC-356 netlist.
8. Released BOM with approved alternates and DNI flags.
9. CPL/pick-and-place file with origin and rotation convention.
10. Fabrication drawing and approved stack-up drawing.
11. Top and rear assembly drawings.
12. Full PCBA STEP model with maximum-height bodies.
13. SWD/programming/test-pad drawing.
14. Manufacturing README with SHA-256 checksums.
15. Bring-up support for FA-001 and FA-002.

## Acceptance gates

No release until:

- Battery and enclosure interfaces are physically reviewed.
- The fabricator approves the exact 0.80 mm stack, minimum drill/pad, USB slots, controlled-impedance geometry, and rush schedule in writing.
- Every part is available, approved, or explicitly consigned.
- Zero unrouted connections remain.
- Zero unresolved ERC/DRC/net-equivalence issues remain.

- PMIC, USB, RF, acoustic, thermal, and manufacturing reviews are signed.
- Output files are regenerated from the final native source and checksummed together.

### Required response from candidate engineer

Confirm:

- Relevant nRF52/nPM1300, USB, RF module, compact wearable, and KiCad experience.
- Who personally performs layout and review.
- Start time and committed 24-36 hour engineering sprint availability.
- Availability for first-board bring-up.
- Fixed or capped estimate and what would stop the sprint.

APPENDIX E

# Mechanical CAD handoff

## Objective

Create a manufacturable rounded-capsule enclosure no larger than 51.0 x 21.0 x 11.0 mm around the released Anticipy PCBA, exact protected battery, 10 mm haptic motor, light pipe, microphone paths, USB-C, button, and chain/bail interface.

## Inputs available now

- Provisional 45.8 x 18.0 mm board-only STEP.
- Provisional board-outline DXF.
- Placement PNG/SVG.
- Electrical requirements and antenna keepout rules.
- Provisional battery and motor envelopes in KiCad.

These are early packaging inputs only. The current STEP exports at 0.82 mm and contains no component bodies. Do not release tooling, aluminum, or final fit from it.

## Inputs required before final release

- Engineer-released PCBA STEP with maximum-height component bodies.
- Exact battery STEP and controlled maximum-tolerance drawing.
- Motor drawing and wire exit.
- Button travel/force and access target.
- USB connector insertion envelope and tolerance.
- LED emitting centre and light-pipe optical requirements.
- Microphone port centres, supplier keepout, mesh and acoustic targets.
- Exact Raytac antenna keepout and polymer-only external zone.

## Material architecture

- Front: 0.60 mm brushed 5052-H32 aluminum.
- Centre: injection-grade polycarbonate geometry; EVT may be printed in polycarbonate if tolerances are verified.
- Rear: 0.60 mm brushed 5052-H32 aluminum.
- Antenna end: polycarbonate only.
- Internal: flame-retardant electrical insulation, battery liner, anti-rattle foam, acoustic mesh, light pipe, controlled adhesive, and strain relief.

## Required CAD outputs

1. ANT-PROD-R0B\_ENCLOSURE\_ASSEMBLY.step
2. ANT-PROD-R0B\_EXPLODED\_ASSEMBLY.step

3. ANT-PROD-R0B\_POLYCARBONATE\_CENTRE.step
4. ANT-PROD-R0B\_POLYCARBONATE\_CENTRE.3mf
5. ANT-PROD-R0B\_FRONT\_ALUMINUM.dxf
6. ANT-PROD-R0B\_REAR\_ALUMINUM.dxf
7. ANT-PROD-R0B\_ANTICIPY\_MARKING.dxf
8. ANT-PROD-R0B\_LIGHT\_PIPE.step
9. ANT-PROD-R0B\_BUTTON\_ACTUATOR.step
10. ANT-PROD-R0B\_MECHANICAL\_DRAWING.pdf
11. ANT-PROD-R0B\_EXPLODED\_ASSEMBLY.pdf
12. ANT-PROD-R0B\_MECHANICAL\_BOM.csv
13. ANT-PROD-R0B\_TOLERANCE\_AND\_ADHESIVE\_SPEC.pdf

## Design checks

- No aluminum, battery, motor, magnet, bail, chain hardware, or conductive coating enters the antenna zone.
- Battery cannot bend, slide, rattle, touch exposed copper, or move toward the antenna after swelling and adhesive creep.
- USB plug reaches full insertion in both orientations without levering the PCB.
- Button is repeatable and cannot remain pressed after a drop.
- Red/blue LED aligns with the light pipe without leaking into the microphones.
- Microphone ports remain clean, short, symmetric, sealed, and protected by acoustic mesh.
- Motor is mechanically isolated enough to avoid unacceptable microphone noise.
- Enclosure can be assembled and reopened during EVT without destroying the PCBA.
- Both sealed first articles pass RF, acoustic, charge, thermal, drop, rattle, and runtime tests.



APPENDIX F

# Firmware handoff

## Status

No production firmware binary is included. This document defines the firmware work needed for ROB bring-up and release.

## Platform

- MCU/radio: nRF52840 in Raytac MDBT50Q-1MV2.
- Power/charger: Nordic nPM1300 over I2C.
- Audio: two PDM microphones on shared clock with separate data source paths.
- Storage: Macronix MX35LF4GE4AD-Z4I 4-Gbit SLC serial NAND.
- Motion: LIS2DW12 over I2C with two interrupts.
- USB: nRF52840 USB 2.0 device through protected D+/D-.
- BLE: secure control and live/stored audio transfer.

## Required behaviors

- Boot safely at the ROB power defaults; raise BUCK1 from 2.7 V to 3.0 V only after the nPM1300 is identified and configured.
- Configure charging only after the exact battery profile and NTC are approved.
- Record each microphone channel, verify channel assignment, and support ambient recording without a button press.
- Implement audio framing, timestamps, corruption recovery, and backfill to iPhone.
- Implement raw NAND ECC, bad-block discovery/management, wear strategy, power-fail recovery, filesystem, secure erase, and near-capacity behavior.
- Support authenticated BLE provisioning, transfer, reset, and firmware update.
- Support USB data and recovery/programming diagnostics.
- Read motion/orientation and motion interrupts.
- Drive red/blue status patterns through nPM1300 LED sinks.
- Drive haptic patterns through the MOSFET and avoid audio corruption/resets.
- Read button events through the nPM1300 interrupt path, not a direct MCU GPIO.
- Expose factory self-tests and measurement logs.

## Required firmware deliverables

1. Source repository and reproducible build instructions.
2. Locked toolchain/SDK versions.
3. Pin-map header matching [ROB\\_PINMAP.md](#).
4. Factory-test image and production image.
5. [.hex](#) or [.bin](#) plus SHA-256 hashes.

6. SWD flash/verify command.
7. Serial-number and provisioning method.
8. Factory test command list and machine-readable pass record.
9. Power-mode table with measured current limits.
10. BLE service/protocol specification.
11. Storage format/recovery specification.
12. Secure reset and firmware-update specification.
13. Known-issues and rollback procedure.

### Bring-up order

1. Power rails and PMIC identity.
2. SWD erase/program/verify and UART logging.
3. I2C scan, nPM1300, and accelerometer.
4. USB enumeration in both cable orientations.
5. BLE advertising, connect, security, and transfer.
6. Microphone capture and channel validation.
7. NAND identity, ECC, bad-block scan, write/read/hash.
8. LED, button, and haptic.
9. Approved battery/NTC and charging.
10. Sealed-unit audio, RF, storage, runtime, and power-fail testing.

APPENDIX G

# PCBA factory handoff

## Status

This folder contains an RFQ and capability package only. It is not a fabrication release.

## Planned build

- 12 four-layer PCBAs.
- Approximately 45.8 x 18.0 x 0.80 mm.
- Components on top side only for one SMT reflow pass.
- Rear battery and haptic motor installed by hand only after programming and bare-PCBA test.
- Build exactly two first articles, then hold the other ten pending written release.

## Files the factory will receive after engineering release

- Release README and checksums.
- Gerber ZIP.
- PTH/NPTH drill ZIP and drill map.
- Fabrication drawing and approved stack-up.
- Released BOM with sourcing and substitution controls.
- CPL/pick-and-place CSV.
- Top and rear assembly drawings.
- Schematic PDF and IPC-356 netlist.
- PCBA STEP.
- Programming/test instructions and firmware hash.
- First-article test plan.

## Required factory confirmations

- 0.80 mm four-layer high-Tg FR-4 is physically in stock.
- 5/5 mil, 0.20/0.45 mm standard through-vias, and two 0.16/0.36 mm USB fanout vias are accepted.
- GCT USB4500 mid-mount slots are accepted.
- Exact 90-ohm USB geometry and panel coupon/TDR will be provided.
- 0201, QFN exposed pad, WSON exposed pad, one top-side stencil, and one reflow pass are supported.
- AOI and X-ray of exposed-pad devices are available.
- No substitution or CAM edit occurs without written approval.
- Two first articles can be assembled and the remaining ten held.

APPENDIX H

# Final assembly SOP

Status: planning document only. Final adhesive, torque, cure, acoustic, and test values must come from the released mechanical package.

## Inputs per unit

- One passed and serialized PCBA.
- One approved protected three-wire battery.
- One tested haptic motor.
- One polycarbonate centre.
- One front and one rear aluminum face.
- Light pipe and button actuator.
- Acoustic mesh for both microphones.
- Electrical insulation/battery liner.
- Anti-rattle foam and strain relief.
- Released adhesives and cleaning materials.
- Bail/chain hardware and packaging.

## Assembly order

1. Verify the serial number, PCBA test record, firmware hash, battery lot, motor lot, and enclosure revision.
2. Inspect microphones, USB connector, antenna region, pogo pads, and solder joints. Quarantine contamination or damage.
3. Program and electrically test through the rear pogo fixture while BT1 and M1 are absent.
4. Measure battery polarity and NTC. Hand-solder BT1 in the frozen order: pin 1 VBAT, pin 2 NTC, pin 3 GND.
5. Add released insulation and strain relief. Confirm the battery cannot touch exposed copper or move toward the antenna.
6. Hand-solder M1 with documented polarity. Add strain relief and mechanical isolation.
7. Retest power, USB, BLE, microphones, storage, motion, LED, button, and haptic before closing.
8. Install the PCBA/battery/motor subassembly into the polycarbonate centre without forcing or bending.
9. Install acoustic mesh, light pipe, button actuator, foam, and seals. Keep adhesive and debris away from microphone ports, USB, and antenna zone.
10. Attach front and rear aluminum faces using the released adhesive pattern and cure process.
11. Run shake/rattle, visual-gap, USB insertion, button, light-pipe, microphone, BLE, charge, thermal, storage/backfill, and runtime tests.
12. Record every result against the serial number. Clean and package only passed units.

## Stop conditions

Stop for smoke, odor, swelling, heat, unstable current, incorrect polarity, damaged microphone, short, programming failure, reset, data corruption, RF loss, audio blockage, rattle, adhesive intrusion, or any unrecorded deviation.



APPENDIX I

# Release and QA gates

## Gate 0 - engineering release

- [ ] Exact battery MPN, drawing, STEP, protection, NTC, polarity, compliance evidence, sample, and stock approved.
- [ ] Every symbol, pin map, footprint, polarity, height, and acoustic port reviewed.
- [ ] PMIC loops and grounds reviewed against Nordic reference.
- [ ] Fabricator returns and Anticipy approves the 0.80 mm stack-up and 90-ohm USB geometry.
- [ ] 0.80/0.82 mm thickness discrepancy closed.
- [ ] Zero unconnected items.
- [ ] Zero unresolved ERC, DRC, or net-equivalence issues.
- [ ] Final PCBA STEP passes enclosure/battery/motor/USB/button/LED/microphone/antenna collision review.
- [ ] Released BOM, CPL, Gerbers, drills, IPC-356, fab/assembly drawings, STEP, test files, and checksums agree.

## Gate A - release remaining ten PCBAs

Both FA-001 and FA-002 must pass:

- [ ] Revision, checksum, material, dimensions, electrical test, AOI, and X-ray evidence.
- [ ] Unpowered short/continuity/isolation checks.
- [ ] Current-limited power-up and all rail measurements.
- [ ] SWD program/verify, device ID, serial, reset, UART, I2C, PMIC interrupt, and button.
- [ ] USB power/data in both orientations.
- [ ] BLE advertising, security, control, and transfer.
- [ ] Both microphone channels, quiet/tone/speech, BLE coexistence, and haptic-noise test.
- [ ] NAND identity, ECC, bad blocks, write/read/hash, power-fail recovery, secure erase, and near-capacity behavior.
- [ ] LED, haptic, accelerometer, and approved battery/charge tests.
- [ ] No open deviation or missing measurement.

## Gate B - sealed-unit release

Both sealed first articles must pass:

- [ ] Fit and retention with no collision, rattle, chafe, or exposed battery risk.
- [ ] Microphone acoustic response and contamination inspection.
- [ ] USB insertion, button travel, and light-pipe alignment.
- [ ] Sealed-device BLE range/RSSI/packet-loss limits.
- [ ] Charge, thermal, drop, shake, overnight recording, backfill, secure reset, and runtime tests.
- [ ] Complete traceability: PCB lot, battery lot, motor lot, firmware hash, enclosure revision, serial, and test record.

## Approval rule

Verbal approval is not release. The approval must identify the exact revision, file checksums, two passing first-article serials, known deviations, and approver.

APPENDIX J

# Supplier contacts

Verified on 2026-08-29 from official company pages.

| Organization                  | Use                                      | Contact                                                                                                                                                                                      |
|-------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CCI Canadian Circuits, Surrey | Primary rush PCB/PCBA capability request | <a href="mailto:sales@canadiancircuits.com">sales@canadiancircuits.com</a> , 604-599-8600, <a href="https://www.canadiancircuits.com/contact/">https://www.canadiancircuits.com/contact/</a> |
| Enigma Interconnect, Burnaby  | Local bare-board backup                  | <a href="mailto:boards@enigmacorp.com">boards@enigmacorp.com</a> , 604-420-3313, <a href="https://www.enigmacorp.com/quote">https://www.enigmacorp.com/quote</a>                             |
| MistyWest, Vancouver          | Full-service hardware engineering option | <a href="mailto:contact@mistywest.com">contact@mistywest.com</a> , 604-292-7036, <a href="https://www.mistywest.com/contact/">https://www.mistywest.com/contact/</a>                         |

These contacts are leads, not booked commitments. Ask for written confirmation of scope, owner, start time, price, material stock, and exact delivery date.